Sahaj Bhadja

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Summary

Full-stack software engineer with 2+ years building **distributed backend APIs**, **search/discovery-adjacent user experiences**, and **cloud-native production systems** using Python, SQL, Kubernetes, and AWS.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 - Present*

- Designed and shipped **backend services and APIs** for 500+ users, reducing response latency from **1.2s to 350ms** through query tuning and distributed service optimization.
- Implemented **fault-tolerant orchestration** for 5K+ monthly jobs with validation and rollback safeguards, improving uptime, reliability, and incident response confidence.

Grad8 **LA, California**
Software Engineer Intern *June 2025 - August 2025*

- Built **full-stack product features** supporting 10K+ daily requests, translating business scenarios into clear, testable APIs and reusable frontend flows.
- Improved **CI/CD and quality guardrails** through automated checks and issue triage, reducing release turnaround by **60%** while preserving product stability.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 - May 2025*

- Built **large-scale analysis pipelines** over 100K+ trajectories, enabling reproducible experimentation and robust ranking/behavior evaluation in complex datasets.
- Produced **debugging diagnostics and technical summaries** that accelerated cross-team decision-making and reduced ambiguity in implementation tradeoffs.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 - July 2024*

- Engineered **distributed API/ETL services** using Python/Java, AWS Glue, and S3, improving response time by **30%** across high-volume data workflows.
- Maintained **20+ automated test suites** with Jenkins, PyTest, and Mockito, increasing pipeline efficiency by **68%** and reducing regression defects.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 - August 2023*

- Developed **MERN + WebSocket modules** for eight workflows and 2M+ synchronized records, improving user discovery and real-time engagement responsiveness.
- Automated **containerized deployments** via Docker/Jenkins, reducing rollout time by **50%** and improving reliability across staging and production paths.

Skills

Languages: **Python**, Kotlin/Java familiarity, JavaScript/TypeScript, SQL, C++
Backend: distributed APIs, service orchestration, MySQL/PostgreSQL, ranking/search-adjacent data flows
Frontend: React.js, responsive UI, discovery and engagement-focused product interfaces
Cloud & Infra: AWS, Kubernetes, Docker, GitHub Actions, CI/CD, observability and reliability practices
Engineering: code reviews, production debugging, cross-functional collaboration, ownership mindset

Projects

Merchant Discovery Search Service | *Python, FastAPI, MySQL, Redis*

- Built a **search and ranking API prototype** for merchant/category discovery with relevance tuning, improving retrieval quality and reducing response latency for user-facing queries.

FinNexus Live | *Next.js, TypeScript, Kubernetes, MongoDB*

- Developed a **production-grade financial platform** serving 50K+ users and 1M+ transactions with scalable APIs, secure integrations, and resilient cloud deployment workflows.

Chat2SQL Code | *Java, TypeScript, SQL, Docker*

- Implemented a **schema-aware query assistant** across 10+ datasets with validation and explainable outputs, improving developer velocity and reducing query debugging overhead.